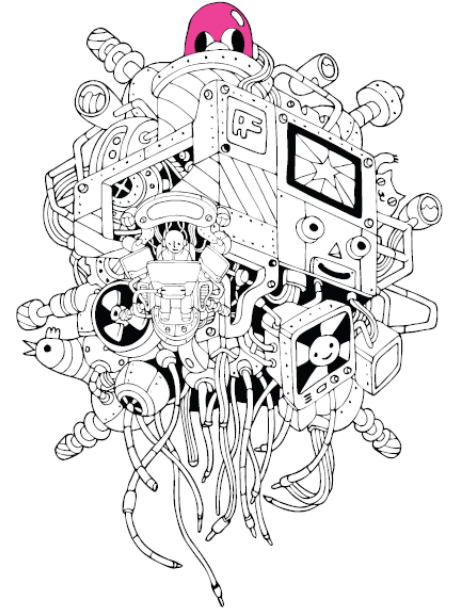


C programming



Chapter 01. This is the C language .

[illegible]

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C language is a programming language .

✓ What is a programming language ?

A programming language is a formal language that can be understood by humans and processed by a compiler or interpreter. C is one of the programming languages.

✓ What about the compiler ?

A compiler translates a program written in a programming language into **machine language** that a computer can understand.

✓ What is Machine Language ?

Machine language is a language composed of 0s and 1s that computers can directly understand and execute.

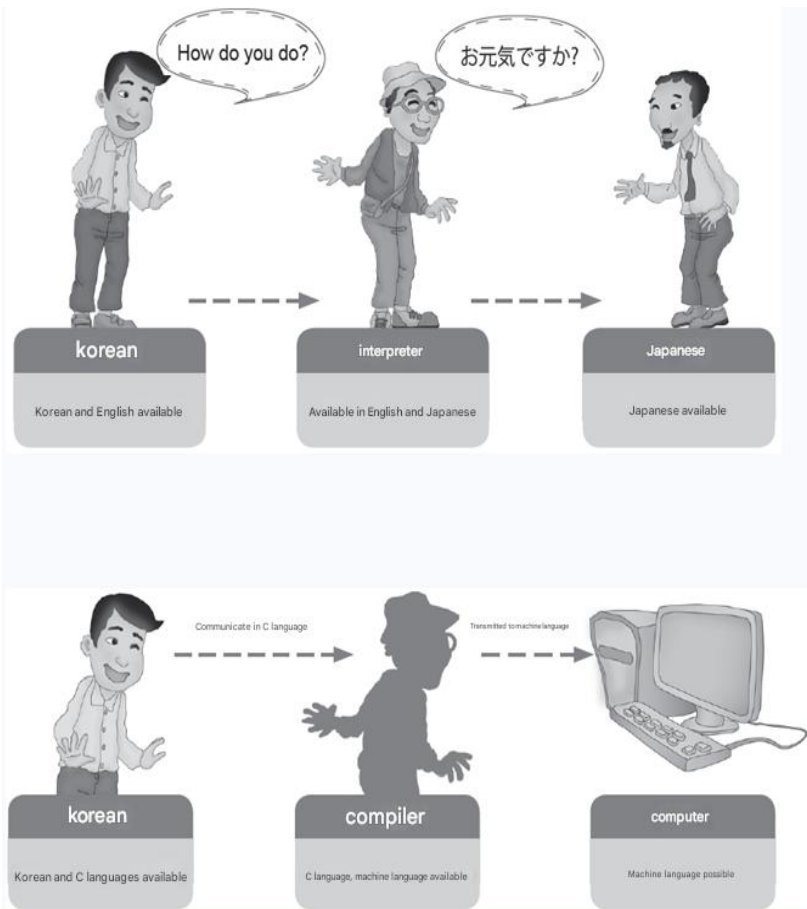
✓ What does it mean to study C language ?

Understanding its grammar and syntax.

Improving your ability to express logical ideas through code.

The more you practice, the more proficient you become.

The more you study others' code, the better your programming skills become.



History and Features of C Language

✓ History of the C language

C is a high-level programming language. It was designed by Dennis Ritchie around 1971 at Bell Labs. It was created to develop the UNIX operating system.

ALGOL 60 (1960) → CPL (1963) → BCPL (1969) → B language (1970) → C language (1971)

✓ Development of UNIX Before C

Before C, UNIX was written in assembly language.

Assembly language depends on the hardware. So UNIX had to be rewritten for each different CPU.

✓ Development of UNIX After C

After C was created, UNIX was rewritten in C.

C is more portable than assembly language. So UNIX did not need to be rewritten for each CPU. It became easier to move UNIX to different computers.

✓ High-level language & Low-level language

A high-level language is easy for humans to read and write.

A low-level language is closer to machine language.

C is mainly a high-level language. But it also has some low-level features because it allows direct access to memory.



Advantages of the C Language

✓ C has procedural features

C is a procedural language. It is easy to learn step by step. Its structure is similar to the way people think.

✓ C programs are portable.

Programs written in C can run on many different computers. You usually do not need to rewrite the whole program for each CPU. However, today there are other languages with better portability than C. So portability is not the biggest advantage of C anymore.

✓ C programs have good performance

Programs written in C usually run fast. They use memory efficiently.

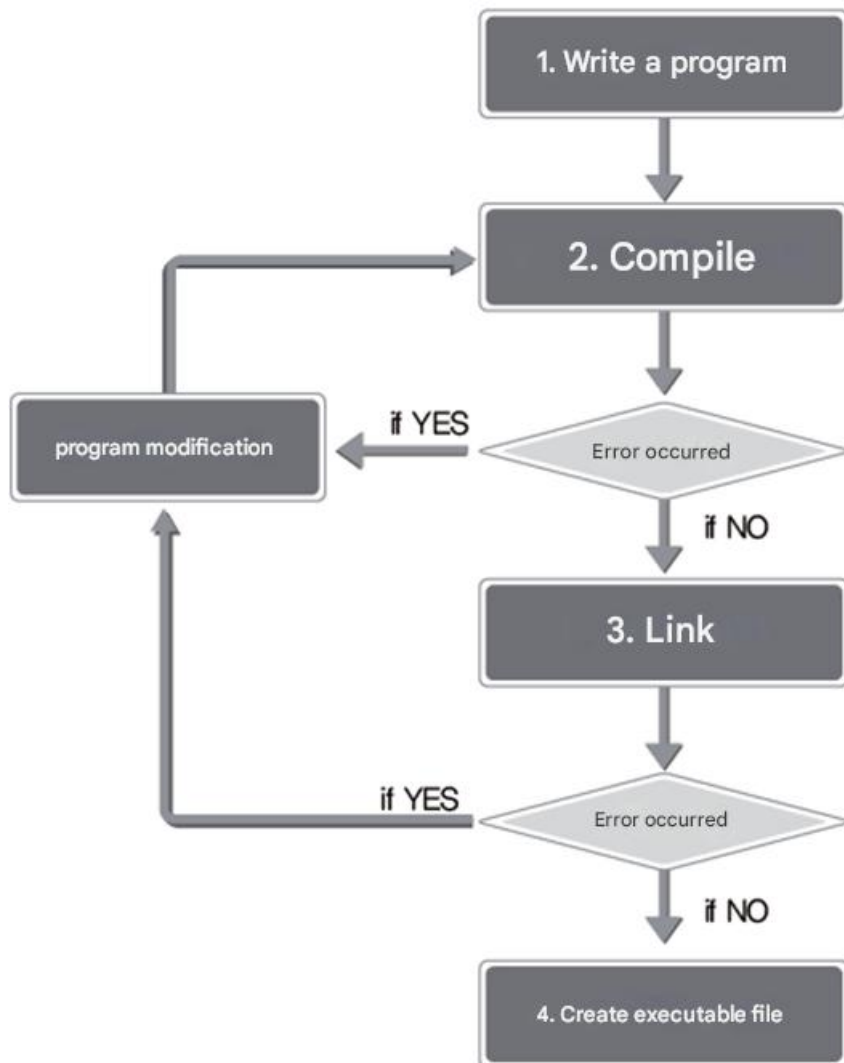
But if the program is written poorly, the performance can become worse.



[illegible]

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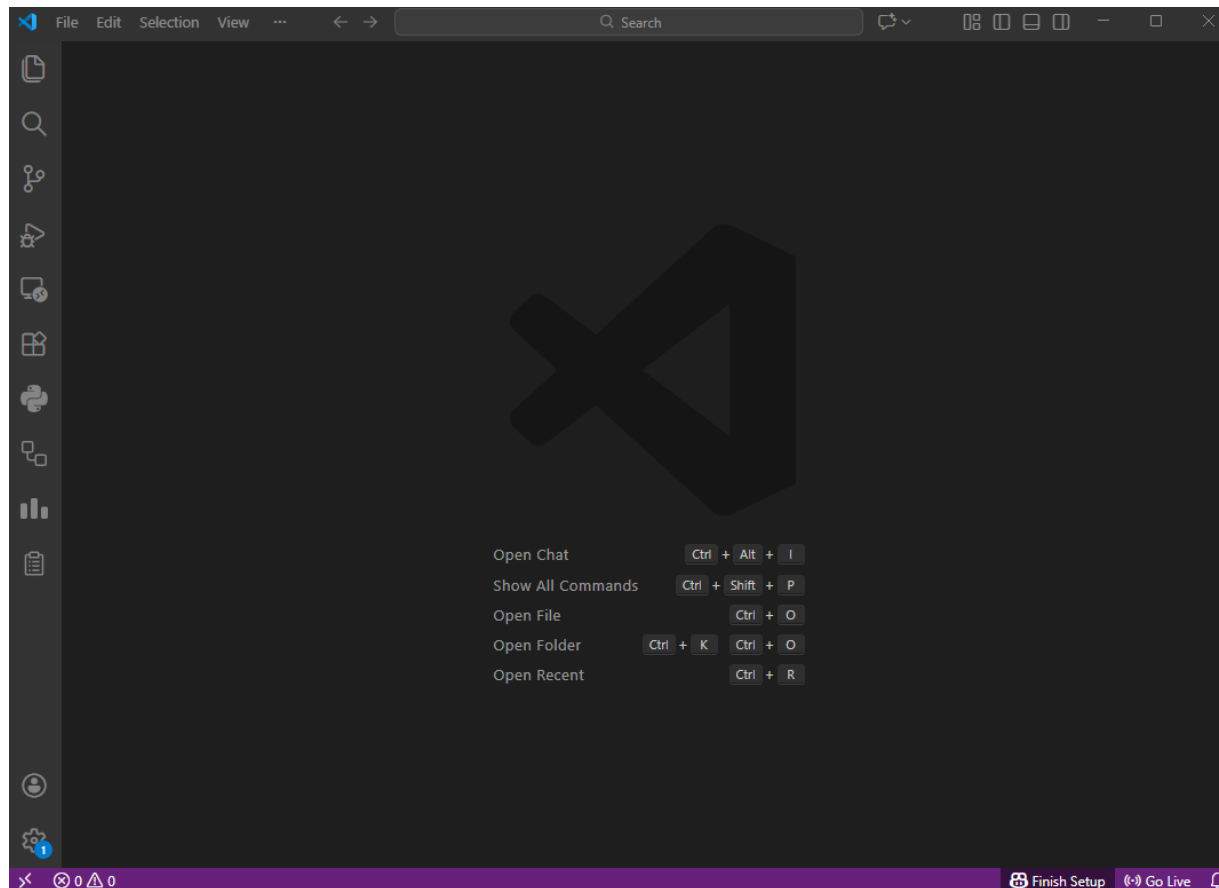
C program completion process (Refer to Week02)



- First step : **writing the program**
- Second step : **Compiling the written program**
- Third step : **Linking the compiled results**

VSC (Execute vsc)

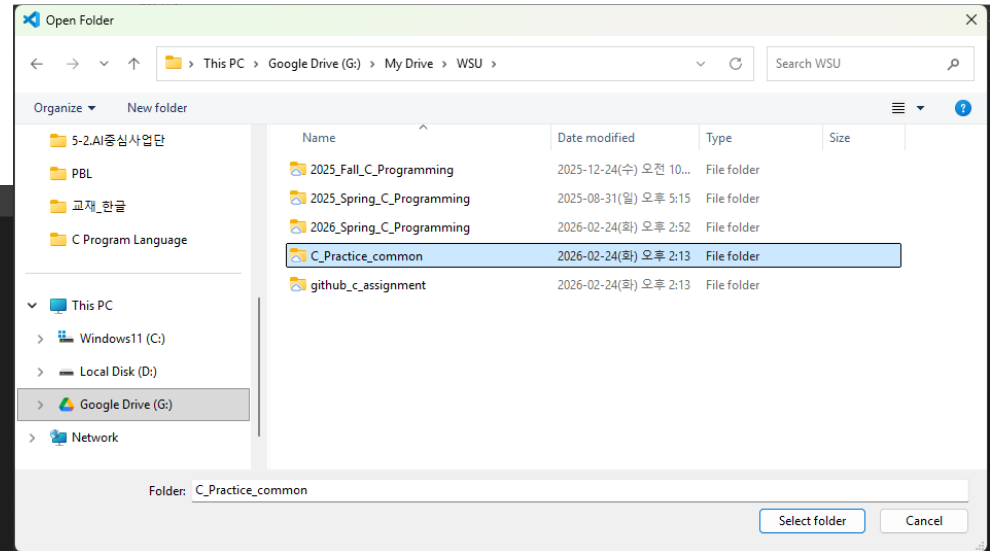
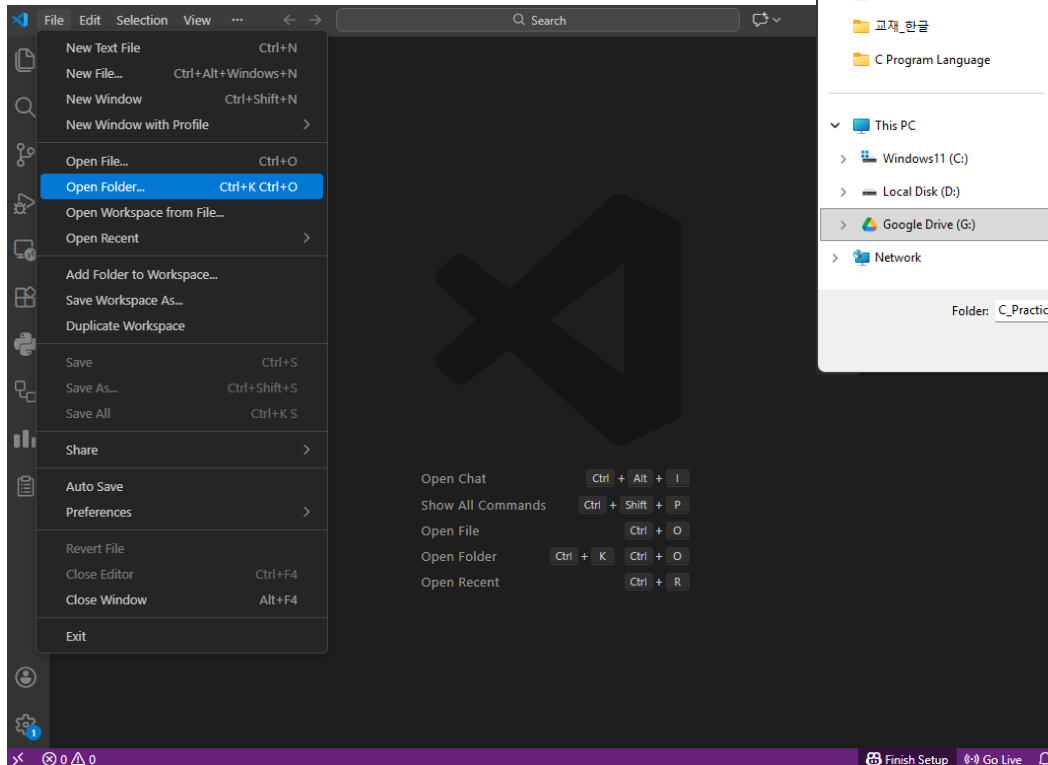
「Running visual studio code)」



VSC (Open Folder)

「Always “Open Folder” to start」

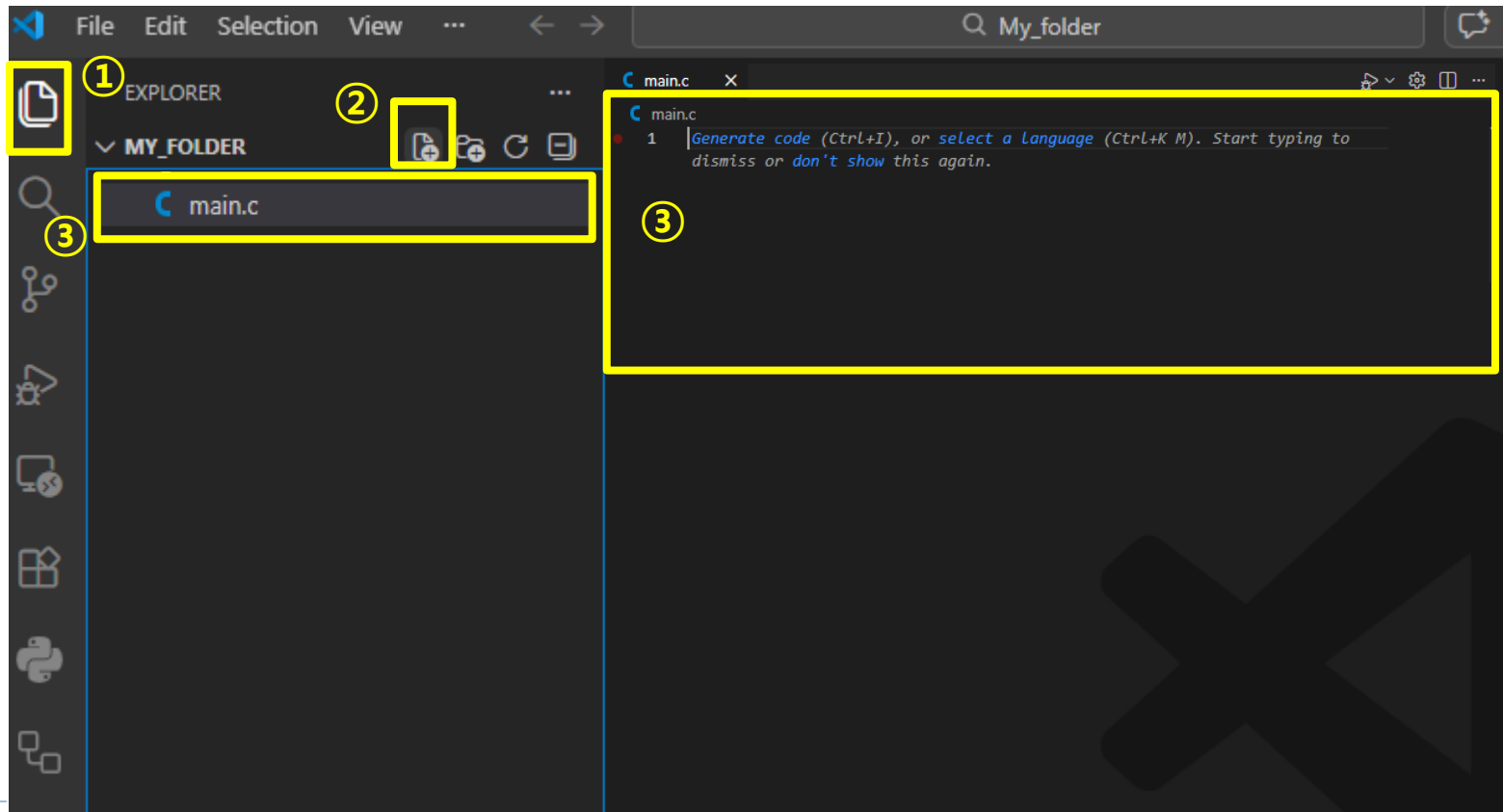
File → Open folder → Select folder



VSC 3 (Create a file)

「 After selecting “Folder” 」

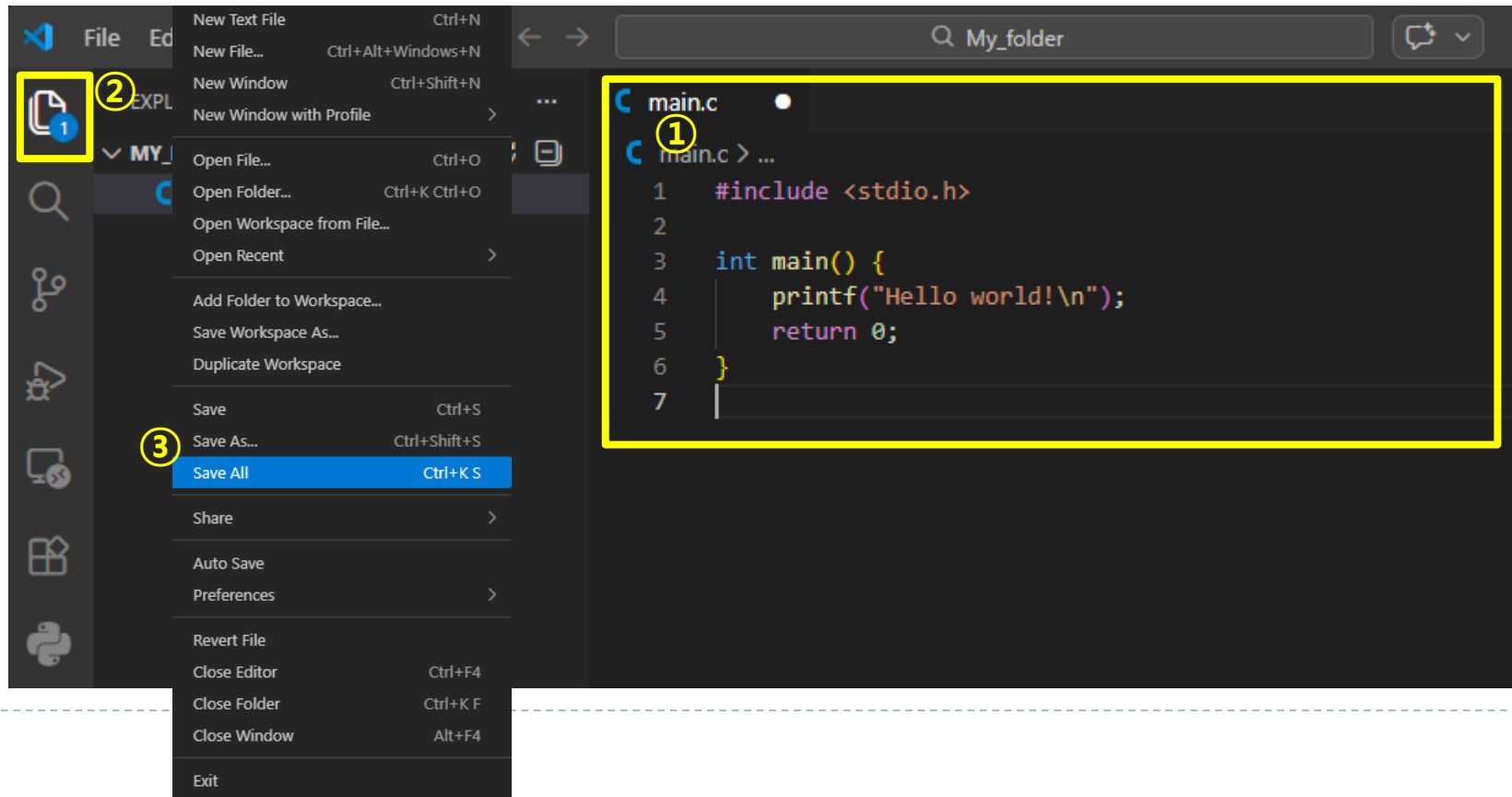
1. Select “Explorer” 2. Click “New file” icon 3. Add filename (**.c) 4. Create a file



VSC 3 (Write & Save a file)

「 Write & Save a file 」

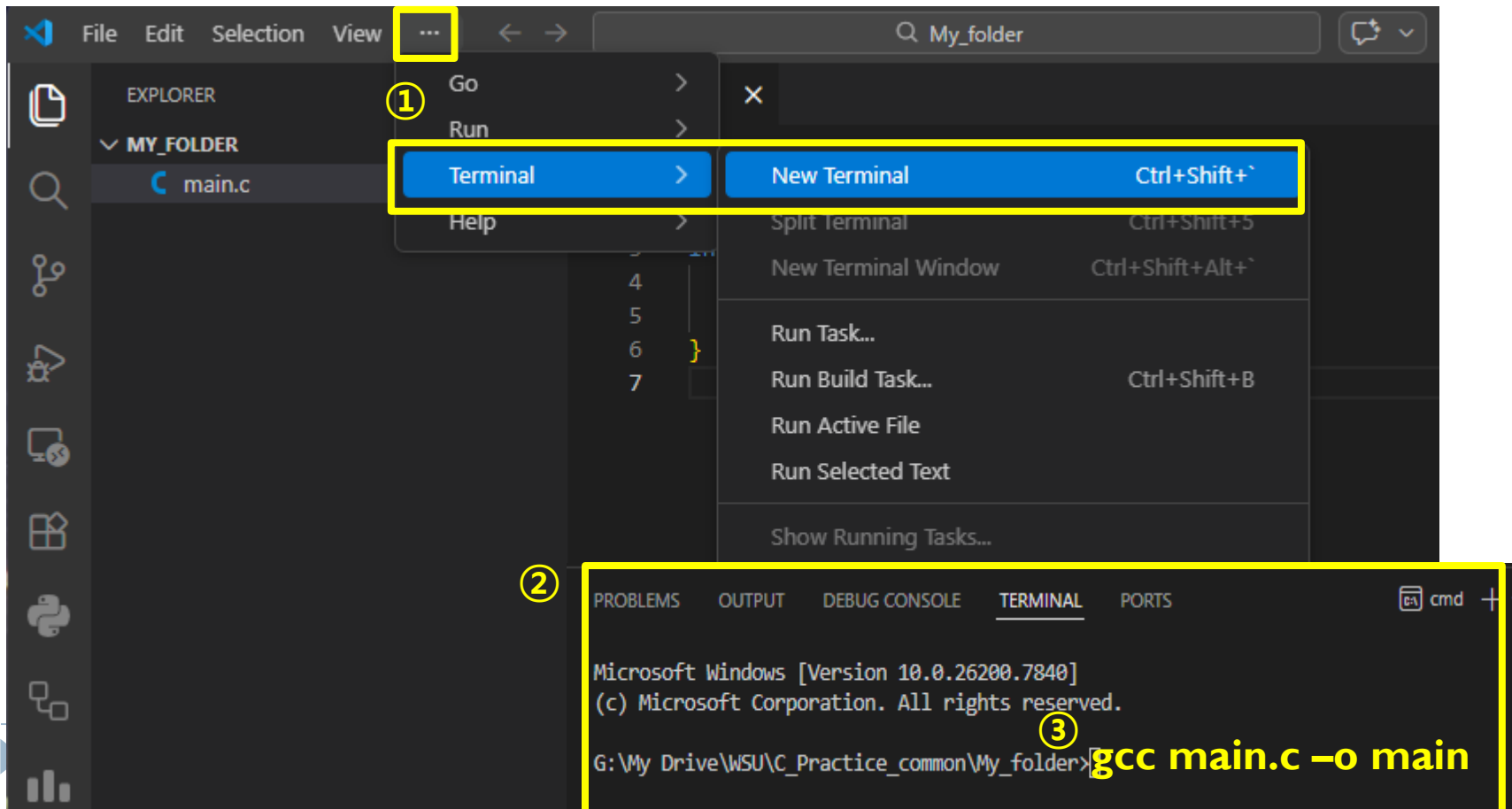
1. Write coding in empty area
2. Check left-top the number tag of "Explorer icon"
3. Save files (File → Save All or Ctrl + s)

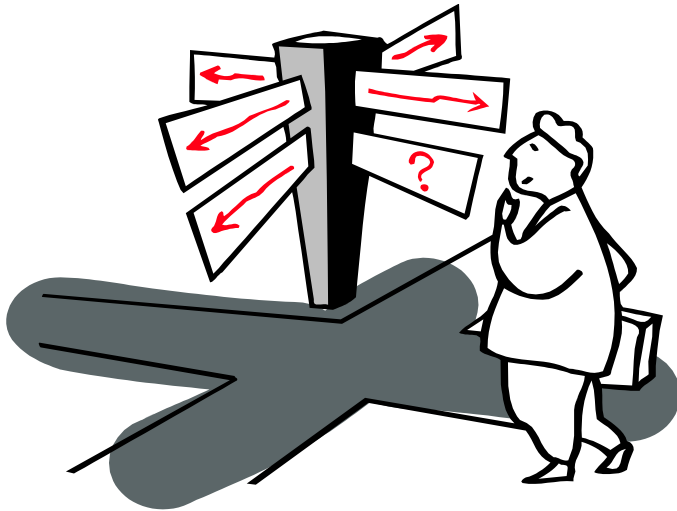


VSC 3 (Build an executive file)

「 After opening a Terminal 」

1. Terminal (or ...) → New Terminal
2. Check Terminal window below
3. Build & Run : `gcc main.c -o main.exe` → `main.exe`





Chapter 01 This That's it.
Do you have any questions ?